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```
%EA_HW_3_7 Nouh Shaikh
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```
clc; clear; close all;
```

```
x = [1,2,3,4,5];
```

```
y = [2.2, 2.8, 3.6, 4.5, 5.5];
```

```
n = length(x);
```

```
Z = [ones(1, n); x; 1./x]';
```

```
coefs = (Z' * Z) \ (Z' * y');
```

```
a = coefs(1); b = coefs(2); c = coefs(3);
```

```
fprintf('a = %.4f, b = %.4f, c = %.4f\n', a, b, c);
```

```
x_fit = linspace(min(x), max(x), 100);
```

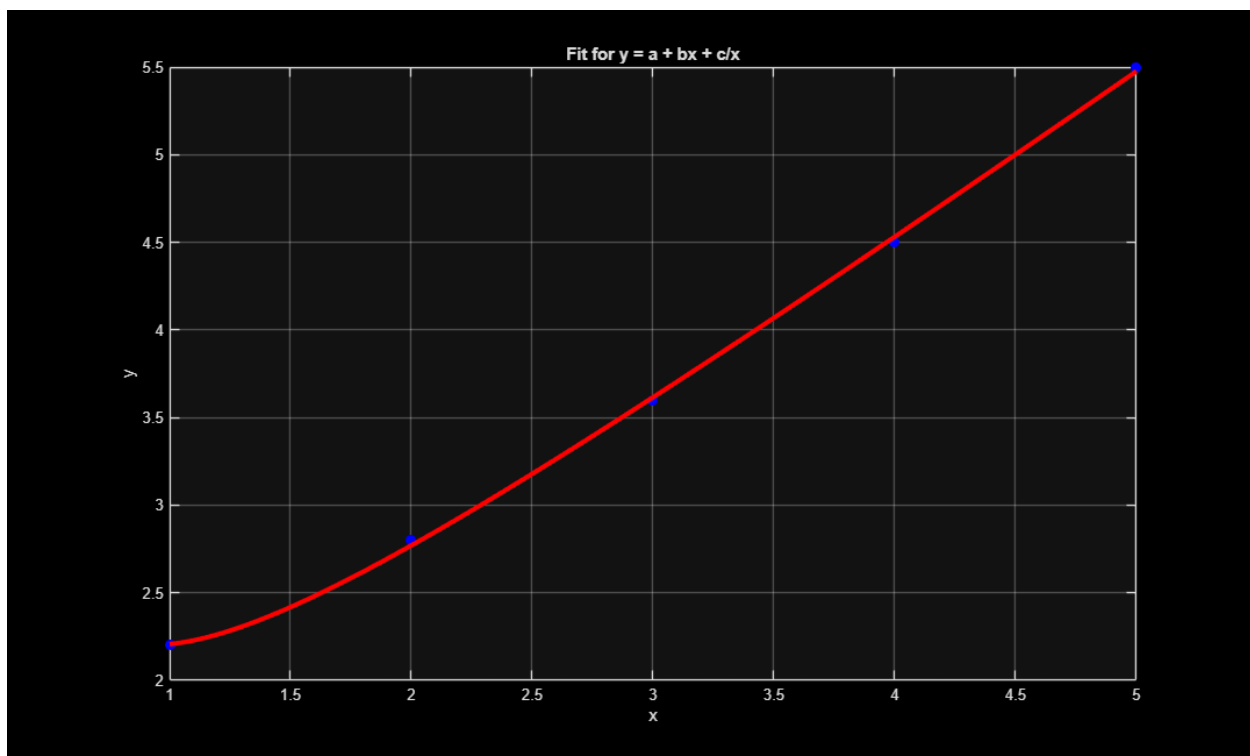
```
y_fit = a + b * x_fit + c ./ x_fit;
```

```
plot(x, y, 'bo', 'MarkerFaceColor', 'b'); hold on;
```

```
plot(x_fit, y_fit, 'r-', 'LineWidth', 3);
```

```
grid on; xlabel('x'); ylabel('y'); title('Fit for y = a + bx + c/x');
```

```
a = 0.3745, b = 0.9864, c = 0.8456
```



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